Using the BOAT design

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BOAT is a Bayesian design for randomized umbrella trials in which the tumor of every enrolled patient gives patient-derived organoids (PDOs). The PDOs are tested with every treatment open in the patient's subgroup, so each patient contributes a clinical response to one treatment and PDO responses to all open treatments. The design

- models patient responses and PDO responses jointly, with a patient random effect shared by both,
- clusters the response mean of each treatment across subgroups by reversible jump MCMC, so that subgroups with similar effects borrow information,
- drops treatments that are not promising at interim analyses and recommends admissible treatments at the end, and
- randomizes each new patient adaptively, favoring the treatment predicted best for this patient from the patient's own PDO responses.

This vignette follows a trial from design to conduct to evaluation by simulation. The MCMC runs below are shortened so that the vignette builds quickly. Use the defaults (10,000 iterations) in practice.

```
library(BOAT)
```

1. Design

The trial below has three subgroups, three experimental treatments and a control. Treatment A_2 is not evaluated in B_1 and A_1 is not evaluated in B_3. Each subgroup enrolls at most 150 patients, with interim analyses after 50% and 75% of them.

```
design <- boat_design(K = 3, J = 3, arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150,
                      interim = c(0.5, 0.75), lambda = 0.70, delta_T = 0.05, delta_R =
                      ↪  0.05,
                      mcmc = list(iter = 1000, burn = 500))
design
#> BOAT design with 3 subgroups and 3 experimental treatments
#>   B1: treatments A0 A1 A3, at most 150 patients
#>   B2: treatments A0 A1 A2 A3, at most 150 patients
#>   B3: treatments A0 A2 A3, at most 150 patients
#>   interim analyses after 50% and 75% of the patients
#>   lambda = 0.70, delta_T = 0.05, delta_R = 0.05
#>   adaptive randomization: gamma = 1, truncation at 0.1
#>   MCMC: 1000 iterations (burn-in 500); randomization model: 2000 iterations (burn-in
↪  1000)
```

A treatment A_j is admissible in subgroup B_k if it is safe, $\pi_{T,j} < \pi_{T,0} + \delta_T$, and effective, $\pi_{R,j,k} > \pi_{R,0,k} + \delta_R$. At an analysis with n_k patients in B_k, a treatment is dropped if its posterior probability of being effective or of being safe is below $c_k = \lambda\sqrt{n_k/N_k}$. At the end the treatments still open are recommended if both probabilities reach λ.

```
bounds <- boat_boundary(design)
bounds
#>    analysis subgroup   n    cutoff
#> 1 Interim 1        1  75 0.4949747
#> 2 Interim 1        2  75 0.4949747
#> 3 Interim 1        3  75 0.4949747
#> 4 Interim 2        1 113 0.6075634
#> 5 Interim 2        2 113 0.6075634
#> 6 Interim 2        3 113 0.6075634
#> 7     Final        1 150 0.7000000
#> 8     Final        2 150 0.7000000
#> 9     Final        3 150 0.7000000
```

2. Conducting a trial

The data of a trial are two data frames. `pat` has one row per patient with the patient `id`, the subgroup `k`, the assigned treatment `j` (0 = control), toxicity `YT` and response `YR`. `pdo` has one row per PDO replicate with the patient `id`, subgroup `k`, the treatment tested `j` and the PDO response `YO`. Here we simulate one trial from assumed true toxicity and response probabilities and keep its data. We then go through its analyses as they would be done during the trial.

```
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
truth
#> True toxicity (piT) and response probabilities (rows k1, k2, ...):
#>       T0   T1  T2   T3
#> piT 0.25 0.35 0.2 0.15
#> k1  0.50 0.30   - 0.70
#> k2  0.50 0.70 0.5 0.30
#> k3  0.30    - 0.5 0.30
trial <- boat_simulate_trial(design, truth, seed = 2031, keep_data = TRUE)
pat <- trial$pat
pdo <- trial$pdo
head(pat)
#>   id k j YT YR
#> 1  1 3 0  0  0
#> 2  2 2 2  0  1
#> 3  3 1 3  0  1
#> 4  4 1 1  0  1
#> 5  5 2 1  1  1
#> 6  6 3 2  0  1
head(pdo)
#>   id k j        YO
#> 1  1 3 0 -1.244765
#> 2  1 3 0 -1.424717
#> 3  1 3 0 -1.064915
#> 4  1 3 0 -1.276883
#> 5  1 3 0 -1.294019
```

```
#> 6  1 3 0 -1.235123
```

The patients are in the order of enrollment. The function below gives the data available once each subgroup B_k has enrolled `n[k]` patients or has been stopped. It also gives the number of patients of each subgroup.

```
data_at <- function(n) {
  keep <- ave(pat$id, pat$k, FUN = seq_along) <= n[pat$k]
  list(pat = pat[keep, ], pdo = pdo[pdo$id %in% pat$id[keep], ], n =
  ↪  tabulate(pat$k[keep], design$K))
}
```

Before the first interim analysis. Each patient is randomized equally among the treatments of the patient's subgroup.

```
boat_randomize(design, k = 1)$prob
#>         0         1         3
#> 0.3333333 0.3333333 0.3333333
```

Interim analysis 1. Fit the model to the data of the first 75 patients of each subgroup and apply the decision rule. `boat_simulate_trial()` runs the MCMC of analysis l with the trial seed plus $1000l$, so the seeds below reproduce the decisions of the simulated trial.

```
d1 <- data_at(bounds$n[bounds$analysis == "Interim 1"])
fit1 <- boat_fit(design, d1$pat, d1$pdo, seed = 2031 + 1000)
fit1
#> BOAT model fit to 225 patients
#>  subgroup treatment  n    pR pr_eff    pT pr_safe
#>        B1        A0 29 0.435     NA 0.224      NA
#>        B1        A1 26 0.369  0.124 0.438  0.0275
#>        B1        A3 20 0.679  1.000 0.123  0.9912
#>        B2        A0 17 0.467     NA 0.224      NA
#>        B2        A1 20 0.642  0.922 0.438  0.0275
#>        B2        A2 23 0.555  0.680 0.273  0.5119
#>        B2        A3 15 0.353  0.010 0.123  0.9912
#>        B3        A0 28 0.333     NA 0.224      NA
#>        B3        A2 19 0.557  0.990 0.273  0.5119
#>        B3        A3 28 0.356  0.294 0.123  0.9912
#> pR, pT: posterior mean response and toxicity probabilities
#> pr_eff: Pr(pi_R,j,k > pi_R,0,k + delta_R | data); pr_safe: Pr(pi_T,j < pi_T,0 +
↪  delta_T | data)
dec1 <- boat_decide(design, fit1, n = d1$n)
dec1
#> Interim analysis
#>  subgroup treatment  n pr_eff pr_safe cutoff decision
#>        B1        A1 75  0.124  0.0275  0.495     drop
#>        B1        A3 75  1.000  0.9912  0.495 continue
#>        B2        A1 75  0.922  0.0275  0.495     drop
#>        B2        A2 75  0.680  0.5119  0.495 continue
#>        B2        A3 75  0.010  0.9912  0.495     drop
#>        B3        A2 75  0.990  0.5119  0.495 continue
#>        B3        A3 75  0.294  0.9912  0.495     drop
#>   B1: open: A0 A3
#>   B2: open: A0 A2
#>   B3: open: A0 A2
```

Randomization until the next analysis. Fit the probit models that predict each patient's response from the patient's PDO responses. Each new patient is then randomized from the patient's mean PDO response to each open treatment. Below we randomize the next patient of the trial.

```
rar1 <- boat_rar_model(design, d1$pat, d1$pdo, active = dec1$active)
i <- max(d1$pat$id) + 1
pat[pat$id == i, ]
#>      id k j YT YR
#> 226 226 3 2  1  0
pdo_mean <- with(pdo[pdo$id == i, ], tapply(YO, j, mean))
pdo_mean
#>           0           2
#> -0.84564077  0.04292128
boat_randomize(design, k = pat$k[pat$id == i], pdo_mean = pdo_mean, rar = rar1)$prob
#>    0    2
#> 0.27 0.73
```

Without a seed, `boat_rar_model()` continues the random number stream of `boat_fit()` as the simulation does. Hence, these are the probabilities with which the simulated trial randomized this patient.

Interim analysis 2. Only the treatments still open after interim analysis 1 are evaluated.

```
d2 <- data_at(bounds$n[bounds$analysis == "Interim 2"])
fit2 <- boat_fit(design, d2$pat, d2$pdo, seed = 2031 + 2000)
dec2 <- boat_decide(design, fit2, n = d2$n, active = dec1$active)
dec2
#> Interim analysis
#>  subgroup treatment   n pr_eff pr_safe cutoff decision
#>        B1        A3 113  1.000   0.998  0.608 continue
#>        B2        A2 113  0.328   0.795  0.608     drop
#>        B3        A2 113  1.000   0.795  0.608 continue
#>   B1: open: A0 A3
#>   B2: accrual stopped
#>   B3: open: A0 A2
rar2 <- boat_rar_model(design, d2$pat, d2$pdo, active = dec2$active)
```

Final analysis. A subgroup stopped at an interim analysis is not reopened, and only the treatments still open are evaluated. `final = TRUE` gives the recommendations.

```
d3 <- data_at(design$N)
fit3 <- boat_fit(design, d3$pat, d3$pdo, seed = 2031 + 3000)
boat_decide(design, fit3, n = d3$n, active = dec2$active, final = TRUE)
#> Final analysis
#>  subgroup treatment   n pr_eff pr_safe cutoff  decision
#>        B1        A3 150      1   1.000    0.7 recommend
#>        B3        A2 150      1   0.806    0.7 recommend
#>   B1: recommend A3
#>   B2: no experimental treatment recommended
#>   B3: recommend A2
```

These decisions agree with those of the simulated trial.

```
trial
#> Simulated BOAT trial (seed 2031): 413 patients
#>   B1: recommend A3; patients per treatment A0 37, A1 26, A3 87
#>   B2: no treatment recommended; patients per treatment A0 31, A1 20, A2 47, A3 15
#>   B3: recommend A2; patients per treatment A0 47, A2 75, A3 28
```

```
#>   accrual stopped early in B2
```

3. Simulation

`boat_get_oc()` simulates many trials with `boat_simulate_trial()` and summarizes them by subgroup and treatment. The run below uses 2 trials with short MCMC runs. In practice, use about 1000 trials.

```
fast <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150, lambda = 0.70,
                    mcmc = list(iter = 300, burn = 150), rar_mcmc = list(iter = 200, burn
                    ↪  = 100))
oc <- boat_get_oc(fast, truth, nsim = 2)
oc
#> Operating characteristics of BOAT from 2 simulated trials
#>
#> B1   (P_k1, P_k2) = (100.0, 100.0)
#>                A0   A1    A3
#> % Final Rec  0.0  0.0 100.0
#> # Patients  35.5 22.0  92.5
#> % Patients  23.7 14.7  61.7
#>
#> B2   FWER_k = 0.0
#>                 A0   A1   A2   A3
#> % Final Rec 100.0  0.0  0.0  0.0
#> # Patients   18.0 19.0 18.5 19.5
#> % Patients   24.0 25.3 24.7 26.0
#>
#> B3   (P_k1, P_k2) = (100.0, 100.0)
#>                A0    A2   A3
#> % Final Rec  0.0 100.0  0.0
#> # Patients  33.0  90.5 26.5
#> % Patients  22.0  60.3 17.7
#>
#> % Final Rec: trials recommending the treatment (A0: no treatment recommended)
#> # Patients, % Patients: mean number and percentage of patients
#> (P_k1, P_k2): power of a subgroup with admissible treatments; FWER_k: of a subgroup
↪  without
```

`boat_dgm()` changes the data-generating values, for example the association ρ between the patient random effect and the PDO responses or the number of PDO replicates M.

The tuning parameter λ is calibrated so that the largest family-wise error rate over the subgroups without an admissible treatment is closest to 10%. Under `truth`, subgroup B_2 has no admissible treatment.

```
cal <- boat_calibrate(design, list(truth), lambda = seq(0.64, 0.76, by = 0.02),
                      nsim = 1000, cores = 8)
cal
```